Stresses in aluminium around an $Al_{13}Fe_4$ inclusion with a given magnetostrictive strain

I. Mikhailovskiy* D. Pshonkin

Moscow Polytechnic University, Moscow, Russia

Abstract

Molecular dynamics at 0 K gives stresses around a spherical $Al_{13}Fe_4$ inclusion of radius 2.5 nm inside aluminium, with a given magnetostrictive strain along the magnetic field. Near it the matrix is compressed by 126 MPa along the field and stretched by 75 MPa across it; shear stresses reach 28 MPa. The response is elastic: even tenfold strain forms no dislocations, and a dislocation above the inclusion shifts 0.78 Å and returns on unloading. Compliant, distorted aluminium at the interface makes the atomistic field about 80% of the elastic solution. If this loss vanishes for micron particles (untested), this strain gives elastic stresses above the alloy yield stress in a shell about one radius thick. With the field on, the observed creep increase would need a third of the measured Al–Mg–Si activation volume, or 6–7 times the measured Fe–Al magnetostriction. Why the effect outlasts the field remains unexplained.

Keywords molecular dynamics; magnetostriction; inclusions; internal stress; aluminium alloys

1 Introduction

Al–Mg–Si alloys containing iron-rich inclusions creep faster after they have been held in a static magnetic field (creep is slow deformation under a constant load), and the effect persists after the field is removed [1–4]. One proposed explanation is magnetostriction of the inclusions, the change of their shape in the field. The matrix resists the change, and stresses appear around the inclusion. These stresses can move dislocations, the line defects whose motion carries plastic deformation. Ref. [3] estimates the stress at the inclusion boundary as 147 MPa in a field of 0.7 T. This estimate does not say how the stress is distributed around the particle, how fast it falls with distance, or how much of it acts on dislocations.

Here the stress field around one $Al_{13}Fe_4$ inclusion in aluminium is calculated. The strain of the inclusion is prescribed: molecular dynamics follows the atoms with an interatomic potential, a formula for the energy of interacting atoms, and this potential does not describe magnetism, so the magnetostriction cannot come out of the calculation itself. Its magnitude corresponds to the 147 MPa estimate [3]. The atomistic result is compared with the elastic solution, and the field is then used to estimate the force on dislocations and the increase in creep rate.

2 Model and method

2.1 The cell

The cell is a periodic box of aluminium, $12.6 \times 12.4 \times 12.6$ nm. A spherical $Al_{13}Fe_4$ inclusion of radius $R = 2.5$ nm sits at its centre (Fig. 1a). The inclusion is cut from an $Al_{13}Fe_4$ crystal with the lattice of the crystal-structure data [5]; aluminium atoms closer than 2.2 Å to an inclusion

*Corresponding author. ilyamihailovsy@gmail.com

atom are removed. The cell holds 118 861 atoms, 114 374 in the matrix and 4 487 in the inclusion. Aluminium surrounds the inclusion on all sides. The boundary conditions are periodic along all three axes, so the cell represents a regular array of inclusions 12.6 nm apart along x and z and 12.4 nm along y; their volume fraction is 3.3%.

A sphere was chosen as the simplest compact shape; the actual shape of the inclusions is not reproduced. The stress around a sphere falls with the distance r from its centre as r^{-3} and is small two or three radii away, so a cell of 10^5 atoms is large enough. A sphere also has an exact elastic solution [6, 7] against which the atomistic result can be checked.

The cell axes are $x \parallel [1\bar{1}0]$, $y \parallel [11\bar{2}]$ and $z \parallel [111]$. The (111) planes on which dislocations glide in aluminium are then horizontal, and the slip direction $[1\bar{1}0]$ lies along x. Atoms interact through the second-nearest-neighbour modified embedded-atom potential for Al–Si–Mg–Cu–Fe [8]; the calculations use LAMMPS [9]. The elastic constants of aluminium with this potential, $C_{11} = 110.5$, $C_{12} = 60.9$ and $C_{44} = 28.4$ GPa, are close to the measured ones.

2.2 The given magnetostrictive strain

The magnetic field points along x. Magnetostriction lengthens the inclusion along the field and shortens it across, without changing its volume (Fig. 1b):

$$\varepsilon_{xx} = \varepsilon, \qquad \varepsilon_{yy} = \varepsilon_{zz} = -\varepsilon/2, \qquad \varepsilon = 1.94 \times 10^{-3}. \tag{1}$$

This ε is the 147 MPa estimate of Ref. [3] divided by a Young's modulus of 75.7 GPa; the modulus of aluminium along [111] with the potential used here is 75.9 GPa. Ref. [3] used 70 GPa, with which 147 MPa corresponds to a strain of 2.1×10^{-3}. The magnetostriction of $Al_{13}Fe_4$ inclusions has not been measured, so ε is an input of the calculation. Every stress below is proportional to ε; for another strain ε' it is multiplied by $\varepsilon'/0.194\%$ (by 1.08 for 2.1×10^{-3}).

Each inclusion atom is moved by $\mathbf{u} = \boldsymbol{\varepsilon}(\mathbf{r} - \mathbf{r}_c)$, where $\mathbf{r}$ is the position of the atom before the move, $\mathbf{r}_c$ the inclusion centre and $\boldsymbol{\varepsilon}$ the strain (1). The inclusion atoms then stay fixed, while the aluminium atoms move until the forces on them vanish (energy minimisation by conjugate gradients). This gives the inclusion the shape it is assumed to take inside the matrix under the field. The arrows in Fig. 1b show the forces with which the inclusion acts on the aluminium, taken from the radial stress at the interface in the elastic solution for a single sphere: outward at the ends along the field and inward on top, at the bottom and at the sides. In a real inclusion magnetic forces would maintain this shape, if a magnetostriction of this size exists.

The strain (1) is the strain of the inclusion inside the matrix. A free particle would lengthen more in the same field, because no matrix would hold it back. If 0.194% were the free magnetostriction of an elastic particle, every matrix stress below would be multiplied by $A = \beta k/(1 + \beta(k - 1))$, where k is the ratio of the shear moduli of inclusion and aluminium and $\beta = 0.46$ is a coefficient of the elastic solution for a sphere that depends only on Poisson's ratio (Supplementary S3); for an inclusion two to two and a half times stiffer than aluminium, $A = 0.63$–0.68.

The reference cell, without strain, is prepared once. The inclusion atoms are fixed from the start: with this potential the equilibrium lattice of $Al_{13}Fe_4$ is 8% larger in volume than the measured one, and a free inclusion would swell. The matrix is held at 300 K for 10 ps, cooled to 0.5 K over 10 ps and relaxed at 0 K. The cell size is then adjusted once so that the mean stress in the aluminium far from the inclusion vanishes. The cells with strain $+\varepsilon$ and $-\varepsilon$ are built from the reference cell by moving the inclusion atoms; only the matrix relaxes in them, at fixed cell size. Half the sum of the responses to $+\varepsilon$ and $-\varepsilon$ shows how far the response departs from linear (Supplementary S2).

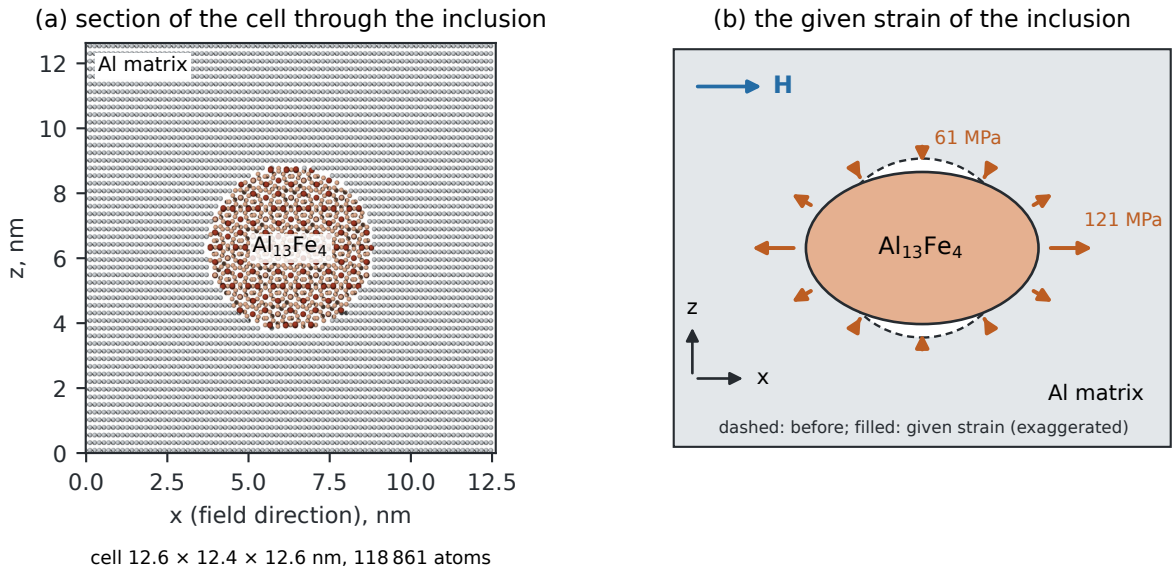


Figure 1: The model. (a) Section of the cell by the xz plane through the inclusion centre, a 1 nm thick slab (OVITO [10]); grey dots are aluminium atoms of the matrix, orange and red ones Al and Fe atoms of the inclusion. (b) The given strain of the inclusion (exaggerated): dashed circle, original shape; filled, shape with the strain (1). Arrows: forces with which the inclusion acts on the aluminium in the single-sphere elastic solution, outward at the ends along the field and inward on top and at the bottom; arrow length is proportional to force per unit surface area, and the numbers give this force along and across the field.

2.3 How the stresses are obtained

The stress at each matrix atom is its virial divided by the atomic volume of aluminium, 16.61 Å^3. The virial of an atom is the sum of the products of the forces from its neighbours and the vectors joining it to them; in a uniform crystal the virials summed and divided by the volume give the ordinary stress. The quantity used is the difference between the strained cell and the reference cell, that is, the stress added by the magnetostriction. Tension is positive and compression negative. Three quantities are used. The normal stress along the radius, σ_{rr}, shows how hard the matrix pushes on the inclusion or pulls it; the 147 MPa estimate is of this kind. The equivalent (von Mises) stress, σ_{eq}, combines all components except the hydrostatic pressure into one number that is compared with the yield stress, the stress at which the alloy starts to deform plastically. The shear stress $\tau = \sigma_{xz}$ acts on the horizontal slip plane along x; it is the stress that pushes an edge dislocation with Burgers vector along x. The Burgers vector is the shift of the crystal left behind when the dislocation passes; for an edge dislocation it is perpendicular to the line.

Values are averaged over atoms in spherical shells 2 Å thick inside narrow cones about chosen directions: along the field (within 10°), across it (within 10° of the plane $x = 0$), and at 25°, 45° and 67° to the field (bands of ±7.5°). The distance $d = r - R$ is measured from the inclusion surface. Apart from the weak elastic anisotropy of aluminium (the dependence of its elasticity on direction) the field is symmetric about the x axis, so for the maps in Fig. 2 the stresses are averaged over rotations about that axis.

2.4 The elastic solution

The atomistic result is compared with two elastic solutions. The first is the exact solution for a single sphere in an infinite isotropic matrix with the same inclusion strain [6, 7, 11]. The second

is obtained for the same periodic cell by a Fourier method [12] with the cubic elastic constants of aluminium from the potential. It includes the neighbouring inclusions of the periodic array and the fixed cell size, so it can be compared with the atomistic result directly. The first shows what the field of a single inclusion in a large grain would be.

3 Results

3.1 How the stress is distributed

Fig. 2 shows how the added stress is distributed around the inclusion. Along the field, at the ends of the inclusion, the matrix is compressed: the inclusion pushes it apart. Across the field the matrix is in tension: the inclusion draws back from it. The shear stress on the slip planes vanishes on the x and z axes and is largest in four inclined lobes, at about 25° and 67° to the field, changing sign from lobe to lobe. The atomistic map and the elastic map for the same cell agree in shape, but the atomistic stresses are about one fifth smaller (Section 3.2). Within 12 Å of the interface the shape differs as well, because a continuum does not describe the atomic structure of the boundary.

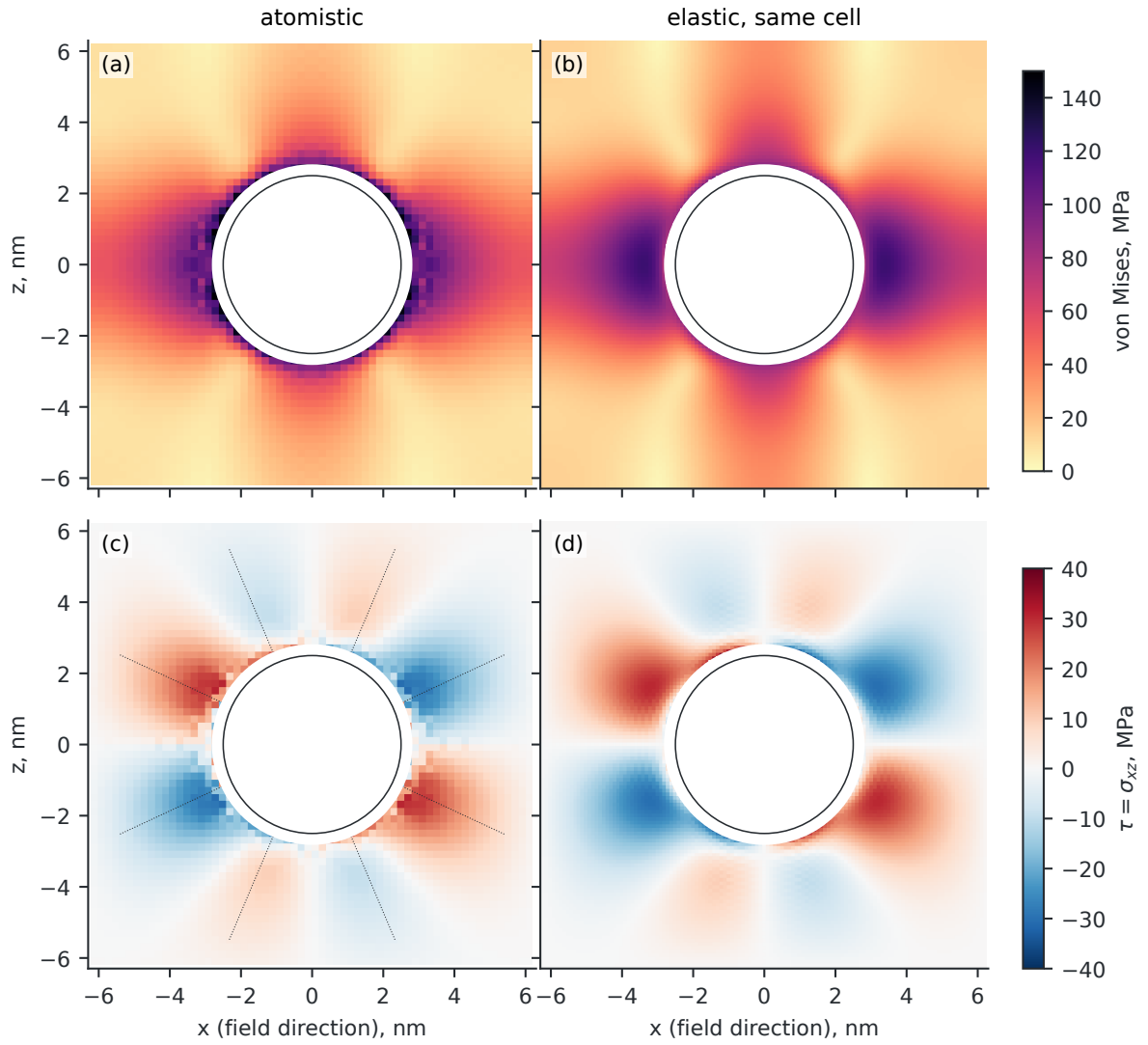


Figure 2: Stresses added by the strain of the inclusion in the xz plane through its centre: equivalent (von Mises) stress (top) and shear stress on the slip plane, $\tau = \sigma_{xz}$ (bottom). Left, atomistic calculation; right, elastic solution for the same periodic cell. The field points along x; the circle is the inclusion surface, and the white ring around it (3 Å of matrix) is left blank. Dotted lines in the lower left panel: directions at 25° and 67° to the field.

3.2 How the stress falls with distance

Fig. 3 gives the profiles along the chosen directions. The atomistic values quoted here are for shells 5–10 Å from the surface; closer in, the narrow cones along the field hold only 15–20 atoms per shell, and the values scatter by up to 25 MPa. Along the field the normal stress is compressive, about 126 MPa, which is close to the 147 MPa estimate of Ref. [3]. The agreement is expected, because ε was derived from that estimate. Across the field the normal stress is tensile, about 75 MPa. At the sides of the inclusion, near the plane $x = 0$, the stress along the field is also large: as the inclusion lengthens, it pulls the aluminium next to it along, and σ_{xx} right at the surface there reaches 121 MPa. The shear stress on the slip planes reaches 28 MPa in the lobes.

One radius from the surface ($r = 2R$) the normal stress along the field falls to about 54 MPa and the shear stress in the lobes to about 19 MPa; two radii away ($r = 3R$) they are about 18 and 7 MPa (single-sphere solution).

From $d \approx 10$ Å on, the atomistic stresses are close to the elastic solution for the same cell

times 0.80; after this scaling they differ from it by 2.5% on average. The factor depends a little on the direction: 0.80 along the field, 0.82 across it, and 0.85 and 0.94 for the shear stress in the 25° and 67° lobes. The atomic displacements, which do not depend on how the stress is computed, are reduced in the same way, and more than 8 Å from the interface the stress of each atom equals the elastic constants of aluminium times its strain to within 3%. The matrix therefore deforms less than in the elastic solution with a perfectly bonded interface. The deficit builds up within 5–8 Å of the interface: in the first aluminium layer the displacement is 94% of the elastic one along the surface normal and 86% along the surface, at 4–6 Å it is already 85% and 73%, and further out it changes little. In this layer the aluminium lattice is distorted, the first layer has half the bulk density, and the atoms carry less stress for the same strain (Supplementary S2).

If the compliance of this layer does not depend on particle size, the weakening scales inversely with the radius and would be about 0.02% for a particle of radius 2.5 μm, so the elastic solution is used below for real particles. This assumption is untested: one inclusion radius and one interface (aluminium atoms within 2.2 Å of the inclusion removed) were computed, at 0 K. Sliding at the interface at service temperature would give a weakening that does not depend on particle size.

The solution for the cell differs from the single-sphere solution: it has the cubic elasticity of aluminium, a fixed cell size and neighbouring inclusions. The shear stress in the lobes differs between them by at most 2 MPa for $3 \leq d \leq 20$ Å, while the normal stresses in the cell are 13–17% larger already at the surface and 30–44% larger at $r = 2R$. The volume fraction of inclusions in the alloy, 0.25%, is more than ten times smaller than in the cell, so the decay far from the inclusion is shown by the single-sphere solution.

The field of a single sphere depends only on r/R, so the elastic solution carries over to inclusions of real size: for a particle 5 μm across ($R = 2.5$ μm) the distances in Fig. 3 are multiplied by a thousand (upper scale). At the given strain the equivalent stress exceeds the yield stress of the alloy, 40–60 MPa [3], out to $r \approx 1.9$–$2.2R$. The largest shear stress on the twelve slip systems (a slip system is a slip plane together with a slip direction in it) exceeds the critical value for slip, 13–20 MPa, out to $r \approx 2.2$–$2.6R$. This critical value is the yield stress divided by the Taylor factor 3.06 [13], which accounts for the different orientations of the grains in a polycrystal. For a 5 μm particle the shell where the equivalent stress exceeds the yield stress is 2.2–3.1 μm thick. Inside this shell the elastic solution overestimates the stresses: in the alloy, dislocations already present would start to move, the matrix would flow plastically and part of the stress would relax. The atomistic calculation does not test this: its matrix is pure aluminium without pre-existing dislocations, and at 0 K no new ones form in it even at a lobe shear stress of 306 MPa (strain 10ε, Supplementary S2).

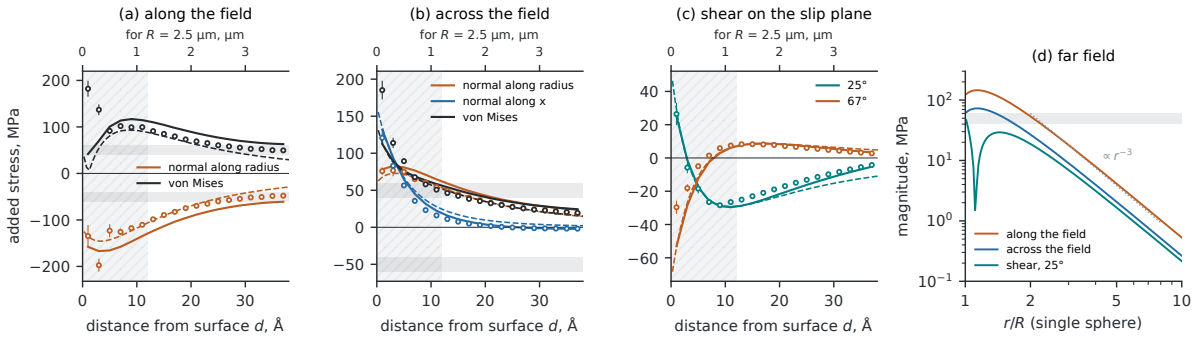


Figure 3: Added stresses against the distance d from the inclusion surface: (a) along the field, (b) across the field, (c) shear stress in the 25° and 67° lobes. Symbols, atomistic calculation (mean over a 2 Å shell and its error); solid lines, elastic solution for the same cell; dashed lines, single sphere; hatched, the layer at the interface where they differ in shape. Away from the interface the symbols lie closer to zero than the solid lines: the atomistic field is about 80% of the same-cell solution. The horizontal band is the yield stress of the alloy, 40–60 MPa. Upper scale: distance for a particle of radius 2.5 μm. (d) Magnitude of the single-sphere stress far from it against r/R on logarithmic scales; dotted, the slope r^{-3}.

3.3 The force on dislocations

A shear stress τ exerts on a dislocation with Burgers vector $b = 2.86$ Å a force τb per unit length, directed along the slip plane (the Peach–Koehler force [14]). The dislocation moves if this force exceeds what holds it back: the lattice resistance (the Peierls stress, the least stress at which a straight dislocation in a pure crystal at 0 K moves to the next position), solute atoms and other dislocations.

Fig. 4 shows the shear stress of the inclusion field averaged along a straight line on which an edge dislocation can lie. The line runs parallel to y at a height h above the inclusion surface. The dislocation is as long as the cell period along y, 12.4 nm. The stress changes sign when x is replaced by $-x$: on one side of the inclusion it pushes the dislocation forward, on the other side back. The averaged stress is largest, 5.8 MPa, at $h = 10$ Å. On the short part of the line right above the inclusion the stress is larger, up to 12.2 MPa, and this part could bow out even when the average is too small to move the whole line. The line average depends on the spacing of the inclusions in the model (12.4 nm along the line) and does not carry over directly to the alloy.

In an Al–Mg–Si alloy dislocations are held by Mg and Si atoms and by precipitates, fine particles of Mg–Si compounds inside the grains; the critical shear stress of the alloy, from its yield stress, is 13–20 MPa. In the single-sphere solution at the given strain the field exceeds this only in a shell 1.2–1.6 radii thick next to the inclusion. Farther out the field is below this value and cannot by itself push dislocations past their obstacles; it can only speed up or slow down their thermally activated motion, in which thermal vibrations of the atoms carry them past the obstacles (see the creep estimate below).

Whether the field moves a dislocation was checked directly. An aluminium slab contains the same held inclusion and a straight edge dislocation on the slip plane 12 Å above it (Supplementary S6). With the inclusion unstrained, the dislocation lies, after energy minimisation, 20.4 Å from the inclusion centre along x, where the line-averaged shear stress of the field is about 4.6 MPa. When the inclusion is given the strain (1), the dislocation moves 0.78 Å away from it, about half the distance $b/2 = 1.43$ Å between neighbouring valleys of the lattice relief (the positions where the energy of the dislocation is lowest), and does not pass into the next valley. When the strain is removed, the dislocation returns to within 0.06 Å of where it was, so in this calculation at 0 K the response is elastic and reversible. The line-averaged field on the dislocation is about half the lattice resistance of the potential, 9 MPa (Supplementary S6). Where the resistance is

lower, for example at room temperature, the same field could move the dislocation further; in the alloy, Mg and Si atoms and precipitates also hold dislocations, and the resistance there is higher (13–20 MPa).

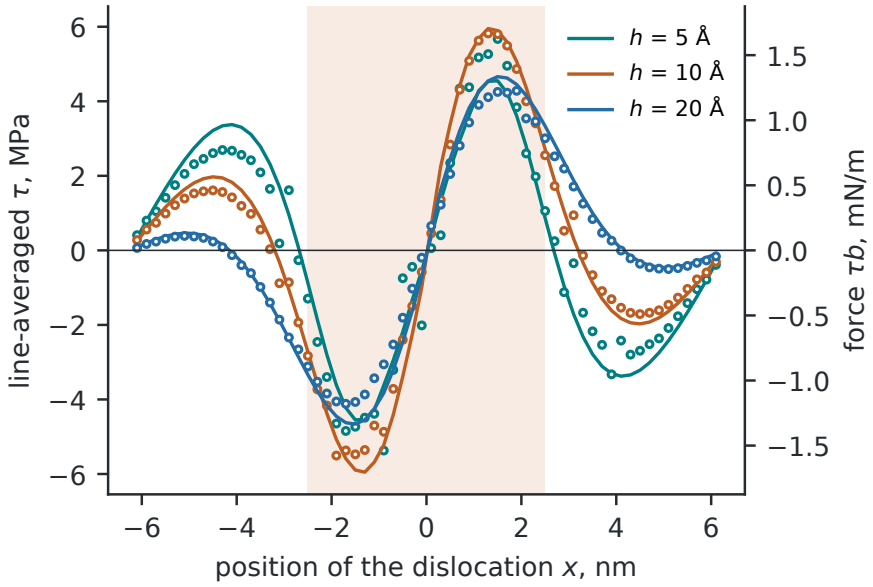


Figure 4: Shear stress averaged along a straight edge dislocation 12.4 nm long that runs above the inclusion at a height h above its surface, against the position x of the dislocation. Symbols, atomistic calculation; lines, elastic solution for the same cell; shaded, the extent of the inclusion along x. Right scale: force per unit length of the dislocation, τb.

3.4 An estimate for creep

The link between the stress field and the creep rate is estimated with a model of thermally activated glide [15, 16]. The estimate applies to creep while the field is on and the stresses around the inclusions exist; the observed increase was measured after the field was removed (see the Discussion). A dislocation passes its obstacles with the help of thermal fluctuations. A shear stress τ makes forward jumps $\exp(V^*\tau/k_\mathrm{B}T)$ times more likely and backward jumps less likely by the same factor; V^* is the activation volume, a measure of how strongly the barrier depends on stress, and $k_\mathrm{B}T$ is the thermal energy. If dislocations around an inclusion meet stresses of both signs equally often, their mean velocity grows by the factor $\cosh(V^*\tau/k_\mathrm{B}T)$, the mean of the factors $\exp(V^*\tau/k_\mathrm{B}T)$ and $\exp(-V^*\tau/k_\mathrm{B}T)$ for the two signs. The excess of this factor over one, integrated over the matrix around one inclusion and multiplied by the number of inclusions per unit volume, f/V_p, gives the relative increase in creep rate (derived in Supplementary S5),

$$H = \frac{f}{V_p}\int_{r>R}\left\langle \cosh\frac{V^*\tau_s(\mathbf{r})}{k_\mathrm{B}T} - 1\right\rangle \mathrm{d}V, \tag{2}$$

where f is the volume fraction of inclusions, V_p the volume of one inclusion, τ_s the shear stress on slip system s (a slip plane together with a slip direction in it), and the angle brackets an average over the twelve slip systems and over grain orientations. The elastic field of a single sphere goes into the integral (the atomistic weakening is taken as negligible for micron-size particles, Section 3.2); it depends only on r/R, so H does not depend on particle size. The values used are $T = 300$ K and $f = 0.246\%$; the latter corresponds to 0.35 wt% of inclusions [3].

In Fig. 5 the solid curve shows the amplitude τ_m needed for the 25% increase in creep rate observed after holding in the field [1] at different V^* if the shear stress falls from the inclusion

surface as $\tau_m(R/r)^3$; this amplitude falls as the activation volume grows. The dashed curve shows the amplitude τ_m for which such a field gives the same increase as the computed field (Supplementary S5). The curves cross at $V^* \approx 23\,b^3$. At $V^* = 70\,b^3$, measured for an Al–Mg–Si alloy [17], the computed field would raise the rate 10^8 times; the model is then outside its range, and a much smaller strain would be enough for the observed 25%. Because H depends only on the product $V^*\varepsilon$, at $V^* = 70\,b^3$ an inclusion strain of about 0.065% is needed. The measured magnetostriction of Fe–Al alloys does not exceed 0.01% [18, 19], six to seven times less.

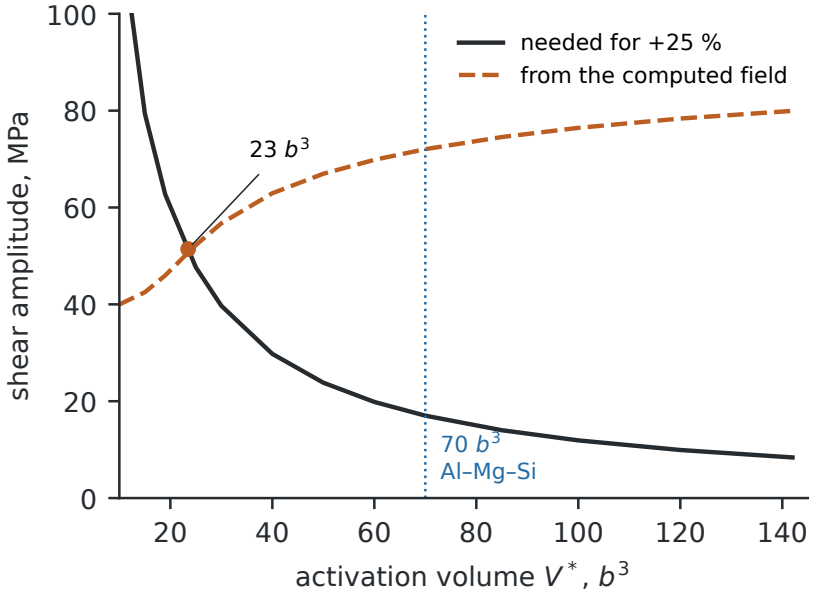


Figure 5: Shear-stress amplitude needed for a 25% increase in creep rate (solid line) and the amplitude equivalent to the elastic single-sphere field of the inclusion with a strain of 0.194% (dashed), against the activation volume V^*; $T = 300$ K, $f = 0.246\%$. Dotted line: $V^* = 70\,b^3$ for Al–Mg–Si [17].

4 Discussion

The calculation gives the stresses that an inclusion of a given shape produces in aluminium. Whether the inclusions take this shape in a field is not known: the magnetostriction of $Al_{13}Fe_4$ inclusions has not been measured, and strong ferromagnetism has not been established for bulk $Al_{13}Fe_4$ [20, 21]. If the real strain of the inclusions is close to the magnetostriction of Fe–Al alloys, the stresses are twenty times smaller than computed, and the shell where they exceed the yield stress disappears. From the single-sphere solution, the largest shear stress on the slip systems at the particle surface exceeds the critical value of the alloy once the inclusion strain is above 0.028–0.043%, and the equivalent stress exceeds the yield stress above 0.045–0.067%. These are strains of the inclusion inside the matrix; the free magnetostriction of an elastic particle would have to be $1/A$ times larger (Section 2.2).

The inclusion is held in its given shape, so its own elasticity does not enter the problem. The potential describes the elasticity of aluminium well but overestimates the lattice of $Al_{13}Fe_4$, which is why the inclusion is held at the measured lattice. The calculation is static (energy minimisation without thermal motion of the atoms) and done at 0 K for one inclusion of one size (radius 2.5 nm) in a regular array; the elastic solution for the same cell includes the field of the neighbouring inclusions. The matrix is pure aluminium without Mg and Si atoms, precipitates or pre-existing dislocations, so the yield stress of the alloy is taken from the literature [3] and

plastic flow near the inclusion is not modelled.

The main limitation concerns the persistence of the effect. In the experiments creep is measured after the field is removed, whereas the elastic stress of magnetostriction disappears with the field. If the field moves dislocations near the inclusions, the displaced dislocations may stay in their new positions after the field is removed; the computed shell where the stress exceeds the yield stress shows where to look for such changes. In the one case computed, a straight dislocation at 0 K, the dislocation returned when the strain was removed; the calculation does not show whether such a rearrangement of defects persists in the alloy or how it affects subsequent creep.

5 Conclusions

An $Al_{13}Fe_4$ inclusion of radius 2.5 nm given a magnetostrictive strain of 0.194% along the field compresses the aluminium matrix along the field by about 126 MPa and stretches it across the field by about 75 MPa. The shear stress on the slip planes reaches 28 MPa in inclined lobes. In the static atomistic calculation at 0 K the matrix responds elastically: no dislocations form even at ten times the strain, and a dislocation above the inclusion is moved by the field by 0.78 Å and returns when the strain is removed. The atomistic field is about 80% of the elastic solution for a perfectly bonded inclusion; the weakening comes from a compliant layer of distorted aluminium at the interface. If its compliance does not depend on particle size, the weakening is negligible for micron-size particles, but one inclusion radius at 0 K does not test this. Far from the inclusion the field falls as $(R/r)^3$; in the elastic solution at the given strain the matrix is stressed beyond the yield stress of the alloy in a shell about one radius thick, 2.2–3.1 μm for a 5 μm particle. In this shell the alloy could deform plastically, that is, by irreversible motion of dislocations already present; this case was not computed atomistically. In a model of thermally activated glide this field, if it acted during creep, would give the observed increase in creep rate for an activation volume of about $23\,b^3$; at the measured $70\,b^3$ a strain of 0.065% would suffice, six to seven times the measured magnetostriction of Fe–Al alloys. The calculation does not explain why the effect persists after the field is removed.

Author contributions

I. Mikhailovskiy developed the method and software, ran the calculations, analysed the results, made the figures and wrote the text. D. Pshonkin conceived the study, supervised it, and reviewed and edited the text.

Data and code availability

The LAMMPS inputs, the programs that build the cell, analyse the results and draw the figures, the reference and strained configurations with atomic stresses, and the result records are available at `https://github.com/iluuua/magnetostriction-plasticity-bounds`.

Conflict of interest

The authors declare no conflict of interest.

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